

Procurement and Inventory Analysis using Generative AI (ChatGPT)

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Business Scenario:

A company manages inventory across multiple materials and suppliers.

Due to limited visibility, there is a risk of stock shortages and inefficient purchasing decisions.

The objective is to use AI to analyze procurement and inventory data and provide recommendations for stock optimization and supplier management.

Approach:

1. Prepared a proof of concept micro-dataset
2. Used Generative AI (ChatGPT) to analyze the data
3. Designed prompts simulating a procurement analyst role
4. Enhanced prompts with SAP S/4HANA context
5. Compared outputs and insights

1. We received the following Excel file with material data:

	A	B	C	D	E
1	Material	Stock	Demand	Supplier	Price
2	chair_01	10	50	A	100
3	table_01	100	40	B	300
4	desk_01	5	20	A	200

2. Go to chat gpt and enter the prompt:

You are a procurement and inventory analyst.

Analyze the following data and:

1. Identify materials with low stock
2. Suggest which materials should be reordered
3. Highlight any risks
4. Suggest possible cost optimizations

Provide a short business summary.

Material Stock Demand Supplier Price

chair_01 10 50 A 100

table_01 100 40 B 300

desk_01 5 20 A 200

3. Check the output:

Here's a structured procurement and inventory analysis based on your data:

1. Materials with Low Stock

Low stock is identified by comparing **Stock vs Demand**:

- **chair_01** → Stock: 10 vs Demand: 50 →  **Critical shortage**
 - **desk_01** → Stock: 5 vs Demand: 20 →  **Critical shortage**
 - **table_01** → Stock: 100 vs Demand: 40 →  **Sufficient / Overstock**
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2. Reorder Recommendations

Immediate Reorder Required

- **chair_01**
 - Shortfall: 40 units
 - Suggested reorder: **50–60 units** (include safety buffer)
- **desk_01**
 - Shortfall: 15 units
 - Suggested reorder: **20–30 units**

No Immediate Reorder

- **table_01**
 - Excess stock: +60 units
 - Action: **Pause ordering**

3. Key Risks

Supply Risk

- Both **chair_01** and **desk_01** rely on **Supplier A**
 -  **Single supplier dependency**
 - Risk of disruption affecting multiple SKUs

Stockout Risk

- Immediate inability to meet demand for:
 - **chair_01**
 - **desk_01**
- Potential consequences:
 - Lost sales
 - Customer dissatisfaction

Overstock Risk

- **table_01**
 - Capital tied up in inventory
 - Storage and holding costs

💰 4. Cost Optimization Opportunities

Inventory Balancing

- Reduce or stop ordering **table_01**
- Reallocate budget to critical shortages

Supplier Strategy

- Diversify away from **Supplier A**
 - Negotiate with **Supplier B** or alternative vendors
 - Reduce dependency risk

Bulk Purchasing

- For **chair_01** (100/unit):
 - Consider bulk discount negotiation
- For **desk_01** (200/unit):
 - Consolidate orders with chairs (same supplier)

Working Capital Efficiency

- Liquidate excess **table_01** stock:
 - Discounts / promotions
 - Bundle with other products

📊 Business Summary

The current inventory shows **critical shortages in chairs and desks**, both sourced from a single supplier, creating a **high supply chain risk** and immediate risk of lost sales. At the same time, **tables are overstocked**, tying up capital inefficiently.

Priority actions:

1. Urgently reorder chairs and desks with safety stock
2. Diversify suppliers to reduce dependency on Supplier A
3. Pause table procurement and reduce excess inventory
4. Negotiate pricing and optimize order quantities

Overall, the business needs to **rebalance inventory**, **mitigate supplier risk**, and **improve capital allocation** to maintain operational stability and profitability.

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4. Add additional context and check the output:

Context:

This data represents materials in a SAP S/4HANA MM environment.

Consider:

- stock levels
- procurement needs
- supplier performance

Here is a refined analysis aligned with an **SAP S/4HANA MM (Materials Management)** perspective, incorporating **stock levels**, **procurement planning**, and **supplier performance**:

1. Stock Level Assessment (MM View)

Critical Materials (Below Reorder Point)

- **chair_01**
 - Stock: 10 | Demand: 50 → **Available-to-Promise (ATP) negative**
 - Status: ● **Stockout imminent**
- **desk_01**
 - Stock: 5 | Demand: 20 → **MRP shortage**
 - Status: ● **Exception message (Reschedule / Expedite)**

Excess Inventory

- **table_01**
 - Stock: 100 | Demand: 40
 - Status: ● **Overstock / Slow-moving**
 - Likely flagged in SAP as **excess stock** or **low turnover**

2. Procurement & MRP Recommendations

MRP Actions (e.g., MD04 / MD02)

- **chair_01**
 - Create **Purchase Requisition (PR)** immediately
 - Suggested lot size: **60 units** (covers demand + safety stock)
 - Consider **lot-sizing procedure** (EX or FX)
- **desk_01**
 - Trigger **planned order** → **convert to PR**
 - Suggested quantity: **25 units**
 - Set **safety stock** to prevent recurrence
- **table_01**
 - Block new PR generation (adjust **MRP type** / **reorder point**)
 - Review **min/max levels**

3. Supplier Performance Insights

Supplier A (chairs, desks)

- Handles **2 critical materials**
- Risks:
 - ⚠ **High dependency (single-source)**
 - ⚠ **Potential bottleneck in case of delays**
- Action:
 - Evaluate **vendor performance (ME61 / KPIs)**:
 - On-time delivery
 - Lead time reliability
 - Price consistency
 - Consider **source list / quota arrangement** to introduce backup vendor

Supplier B (tables)

- Overstock situation suggests:
 - Either **over-ordering** or **forecast inaccuracy**
- Action:
 - Renegotiate **delivery schedules (scheduling agreements)**
 - Shift to **smaller, more frequent deliveries**

4. Key Risks in SAP Context

Operational Risks

- **Stockout** (chair_01, desk_01)
 - Impacts:
 - Production orders (if components)
 - Sales orders (availability check failure)

Planning Risks

- **Inaccurate MRP parameters**
 - Missing or low **safety stock**
 - Incorrect **reorder points**
 - Poor **forecast integration** (MD61)

Supplier Risk

- No **quota arrangement** → no flexibility
- No **info record optimization** → pricing/lead time not leveraged

5. Cost Optimization Opportunities

Inventory Carrying Cost

- Reduce **table_01 stock**
 - Apply:
 - **Stock transfer** (if multi-plant)
 - Promotions / liquidation

Procurement Optimization

- Consolidate orders with **Supplier A**
 - Negotiate:
 - Volume discounts
 - Better payment terms

MRP Parameter Tuning

- Introduce:
 - **Safety stock levels**
 - Proper **lot-sizing** (e.g., EOQ-based)
- Prevent:
 - Emergency procurement (higher costs)

Strategic Sourcing

- Add secondary supplier for:
 - chair_01
 - desk_01
- Use:
 - **Quota arrangement** to split demand

Executive Business Summary

In the current SAP S/4HANA MM setup, the inventory shows **critical shortages in chairs and desks**, driven by insufficient planning parameters and **over-reliance on a single supplier (Supplier A)**. This creates immediate risks to order fulfillment and production continuity.

At the same time, **tables are significantly overstocked**, indicating **inefficient demand planning and excess working capital tied in inventory**.

Key actions:

- Trigger urgent procurement via MRP for shortage materials
- Adjust MRP parameters (safety stock, reorder points)
- Diversify sourcing using quota arrangements
- Reduce excess stock and improve demand forecasting

Overall, improving **MRP configuration, supplier strategy, and inventory balancing** will enhance service levels while reducing cost and risk.

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Conclusion

This project demonstrates how AI can support procurement and inventory decision-making by providing fast and actionable insights.

It highlights the potential of combining SAP knowledge with AI tools to improve business processes.